

**Choosing Between Machine Learning and Deep Learning: Customer Churn and Machine
Translation**

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The lesson explains the difference between the two approaches in a practical way. In traditional machine learning, people choose the important features and an algorithm such as a support vector machine or logistic regression learns from them. In deep learning, a layered neural network learns useful features directly from raw data (Indiana Wesleyan University [IWU], 2026). The main question is not which approach is more advanced. It is which one fits the problem. This report compares customer churn prediction with machine translation and uses three questions to judge the fit: What kind of data is available? How much data and computing does the task require? And do the people using the result need a clear explanation?

The Machine Learning Case: Customer Churn Prediction

The lesson presents churn prediction as a traditional machine learning example. A telecom company can use features such as customer age, contract length, and monthly charges to predict which customers are likely to leave (IWU, 2026). Dao (2022) studied a real telecom churn problem using kernel support vector machines. The study followed a familiar machine learning process: selecting useful features, correcting the imbalance between customers who stayed and customers who left, and tuning the model. The final model reached an F1-score near 99%. This matters because identifying high-risk customers gives the company a chance to respond before losing them, and retaining an existing customer generally costs less than replacing one (Dao, 2022).

Why Classical Machine Learning Fits This Problem

Churn data is usually stored in a structured table, and the columns already have business meaning: tenure, plan type, charges, and support calls. The company defined these variables

when it built its billing and customer systems, so the model does not need to discover them from raw images, audio, or text. This is the kind of problem where classical models usually work well. Research across common tabular datasets has also found that tree-based and other classical models often outperform deep neural networks on datasets of moderate size (Grinsztajn et al., 2022). Many telecom churn datasets contain thousands or tens of thousands of labeled customers rather than millions, and the classes are often imbalanced. That makes careful feature selection and resampling especially useful. Interpretability also matters. If a model shows that a contract nearing its end or repeated support calls increased a customer's risk, the retention team has something specific to act on. Classical models are also faster and less expensive to retrain as customer behavior changes.

Why Deep Learning Is Less Suitable Here

Deep learning is most valuable when it can discover patterns inside raw, high-dimensional data such as pixels, audio, or long text sequences (IWU, 2026). A typical churn table does not contain that kind of raw signal. The company has already defined the features, so there is less benefit in asking a deep network to learn them again. Deep models also require more data, computing, and tuning, while the available churn history may fall in the range where simpler models perform as well or better (Grinsztajn et al., 2022). They can also make the result harder to explain. A retention agent needs to know why a customer was flagged so the response can match the problem. In this case, deep learning adds cost and reduces transparency without a clear reason to expect a better business result. It is not impossible to use, but it is less suitable for this type of structured problem.

The Deep Learning Case: Machine Translation

The lesson identifies natural language processing as a deep learning application because recurrent networks and Transformer models can learn context and meaning from sequences of words (IWU, 2026). Google Translate is a practical real-world example. During its first decade, the service used phrase-based statistical translation, which translated sentences in smaller pieces. In September 2016, Google introduced its Neural Machine Translation system and moved all Chinese-to-English production traffic to it, about 18 million translations per day. Google reported error reductions of 55% to 85% on major language pairs in human evaluations (Le & Schuster, 2016). Because I move between French and English regularly, I understand why this matters. A translation can be technically correct word by word and still sound unnatural or miss the meaning of the sentence.

Why Deep Learning Fits This Problem

Translation is very different from churn prediction. The input is raw text, its length changes from sentence to sentence, and meaning depends on word order, agreement, idioms, and context. These relationships are difficult to reduce to a small list of manually selected features. Deep networks are well suited to this problem because they learn increasingly complex representations from the text instead of depending only on features chosen in advance (IWU, 2026). Google's system reads the full source sentence before producing the translation and uses attention to identify which source words matter for each output word. Google also noted that the neural approach required fewer hand-designed engineering choices than the phrase-based system it replaced (Le & Schuster, 2016). Deep learning's need for large datasets and strong computing resources is a disadvantage in some problems, but translation has web-scale parallel text and the specialized hardware needed for production use.

Why Classical Machine Learning Is Less Suitable Here

Classical models such as support vector machines and logistic regression usually need fixed-length, manually designed features. That is a poor match for sentences because the important information is not only which words appear, but how they relate to one another across the full sequence. Earlier phrase-based statistical systems handled parts of sentences well but often missed broader context. Google's move from phrase-based translation to a neural production system shows why deep learning became the stronger approach for this task (Le & Schuster, 2016). The lesson compares classical machine learning to giving a student a checklist and deep learning to teaching the student how to read (IWU, 2026). Translation is closer to a reading problem because the model must account for context, order, and meaning together. Classical methods can still support parts of the process, but they are less suitable as the main translation approach.

Conclusion

The two examples show that the best method depends on the problem, not on which one sounds more advanced. Three questions help make the choice. Does the data arrive as a structured table or as raw, complex input? Is there enough data and computing power for representation learning? And do the people using the result need to understand why the model made the prediction? Customer churn usually involves structured data, moderate dataset sizes, and a real need for explanation, so traditional machine learning is the better fit (Dao, 2022; Grinsztajn et al., 2022). Machine translation involves raw sequences, large amounts of training data, and complex relationships that are difficult to define manually, so deep learning is more suitable (Le & Schuster, 2016). The main lesson for me is simple: choose the method that fits the data and the decision, not the method that appears most impressive.

AI Use Disclosure

I used Boodle Box (Boodle AI, 2026) to fix my grammar, and check my APA formatting as well as my citations. I selected the examples, evaluated why each approach fit its problem, checked the claims and citations, and revised the final wording so it reflects my own analysis.

References

- Boodle AI. (2026). *Boodle AI* [Large language model]. <https://box.boodle.ai>
- Dao, V. T. S. (2022). Churn prediction in telecommunication industry using kernel support vector machines. *PLOS ONE*, 17(5), Article e0267935.
<https://doi.org/10.1371/journal.pone.0267935>
- Grinsztajn, L., Oyallon, E., & Varoquaux, G. (2022). Why do tree-based models still outperform deep learning on typical tabular data? *Advances in Neural Information Processing Systems*, 35. <https://neurips.cc/virtual/2022/poster/55627>
- Indiana Wesleyan University. (2026). *Machine learning vs. deep learning* [Course lesson]. AIMA-500: Machine Learning Fundamentals (3SU2026), Brightspace.
<https://brightspace.indwes.edu>
- Le, Q. V., & Schuster, M. (2016, September 27). *A neural network for machine translation, at production scale*. Google Research Blog. <https://research.google/blog/a-neural-network-for-machine-translation-at-production-scale/>